

A Meshfree Reproducing-Kernel Kirchhoff–Love Shell in Tahoe

Formulation, Implementation, Boundary Conditions, and Verification
against Wang & Bazilevs (2025)

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September 27, 2026

Technical Report — Tahoe `develop`, issues #59–#73

Abstract

This report documents **RKShellT**, a rotation-free meshfree Kirchhoff–Love (KL) shell element added to the Tahoe finite-element framework. The element follows the reproducing-kernel particle method (RKPM) formulation of Wang and Bazilevs [1]: point-cloud geometry through local principal-component charts, quadratic RK approximation with a cubic B-spline kernel, nodal integration with Taylor-series (natural) stabilization, three-point through-thickness integration of a plane-stress J_2 plasticity model, co-rotational stress update and a thickness update. Two essential-boundary-condition treatments are provided for the non-interpolatory field: direct nodal collocation and a new variational penalty controller that reports the physical reaction. The element is verified against the Belytschko shell obstacle course and validated against the elasto-plastic examples of the reference paper. With the current code, the Scordelis–Lo roof reproduces the paper’s four-mesh convergence to three digits, the necking cylinder follows the paper’s stabilized load curve within 1–4% through the peak and into softening, and the elasto-plastic pinched cylinder lies inside the published band once the published curves are recognized to plot the symmetry-model load, one quarter of the physical point load. Remaining limitations and the test infrastructure are summarized at the end.

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1 Introduction and scope

Thin structures under large deformation, folding and self-contact are a natural target for mesh-free methods: the approximation is smooth, refinement does not need a conforming mesh, and higher-order continuity is available without the C^1 element constructions that finite-element Kirchhoff–Love shells require. Wang and Bazilevs [1] proposed a general-purpose meshfree KL shell built on RKPM, and demonstrated it on the elastic obstacle course, geometric accuracy on a point cloud, necking of a cylindrical shell, and the elasto-plastic pinched cylinder.

This report describes the implementation of that formulation in Tahoe (GitHub epic #59 and sub-issues #60–#73), the boundary-condition machinery that a non-interpolatory field needs, and the evidence that the implementation reproduces the paper. It supersedes the earlier working notes in `docs/meshfree_kl_shell/` where they disagree; those notes are kept for history.

Units follow the reference paper: the elastic benchmarks use the consistent unitless data of the obstacle course, and the elasto-plastic cases use millimetres, megapascals and newtons (density in tonne/mm³).

2 Formulation

2.1 Kinematics

The shell is described by its mid-surface $\bar{\mathbf{x}}(\xi^1, \xi^2)$ and unit normal $\mathbf{n}$; a material point at through-thickness coordinate $\xi^3 \in [-1, 1]$ sits at

$$\mathbf{x}(\xi^1, \xi^2, \xi^3) = \bar{\mathbf{x}}(\xi^1, \xi^2) + \xi^3 \frac{h}{2} \mathbf{n}(\xi^1, \xi^2), \quad \mathbf{n} = \frac{\bar{\mathbf{x}}_{,1} \times \bar{\mathbf{x}}_{,2}}{\|\bar{\mathbf{x}}_{,1} \times \bar{\mathbf{x}}_{,2}\|}. \quad (1)$$

The Kirchhoff–Love hypothesis makes the normal a function of the mid-surface position, so the only unknowns are the three mid-surface displacement components: the element is rotation-free. Membrane strains follow from first derivatives of $\bar{\mathbf{x}}$ and curvatures from second derivatives, so the approximation must be at least C^1 ; the RK basis below is C^2 .

The strain–displacement operators are written with the auxiliary tensors of the reference paper (its Eqs. 39–55): $\mathbf{B}_1, \mathbf{B}_2$ express the variation of the normal through the tangent derivatives, and the spatial velocity gradient at each through-thickness station is formed from the current mid-surface tangents and the inverse of the station’s deformation map. These operators are in `KLShellKernels.h` and are verified against finite differences to machine precision (unit tests, Section 6).

2.2 Reproducing-kernel approximation on a point cloud

No surface parameterization is assumed. For each node a local chart is built by principal component analysis (PCA) of its neighbourhood: the two dominant principal directions span the tangent plane and give local coordinates $\boldsymbol{\xi} = (\xi^1, \xi^2)$. On that chart the RK shape functions are

$$\Psi_I(\boldsymbol{\xi}) = \mathbf{H}^T(\mathbf{0}) \mathbf{M}^{-1}(\boldsymbol{\xi}) \mathbf{H}(\boldsymbol{\xi} - \boldsymbol{\xi}_I) \phi_a(\boldsymbol{\xi} - \boldsymbol{\xi}_I), \quad \mathbf{M}(\boldsymbol{\xi}) = \sum_J \mathbf{H}(\boldsymbol{\xi} - \boldsymbol{\xi}_J) \mathbf{H}^T(\boldsymbol{\xi} - \boldsymbol{\xi}_J) \phi_a(\boldsymbol{\xi} - \boldsymbol{\xi}_J), \quad (2)$$

with $\mathbf{H}$ the monomial basis complete to degree p (quadratic, $p = 2$, as in the paper) and ϕ_a the cubic B-spline kernel of radius $a = s \Delta x$, where s is the normalized support (`support_factor`) and Δx the nodal spacing. The mid-surface displacement is $\mathbf{u}^h(\boldsymbol{\xi}) = \sum_I \Psi_I(\boldsymbol{\xi}) \mathbf{d}_I$. The shape functions and their first, second and third derivatives come from Tahoe’s existing meshfree core (MLSSolverT with CubicSplineWindowT); issues #61, #64 and #69 fixed the on-node limits

of the spline derivatives and the diagonal second derivatives that this element exercises for the first time.

Two consequences matter for users.

- The field is non-interpolatory: the coefficient $\mathbf{d}_I$ is not the displacement of node I . Displacement output, boundary conditions and reactions must be handled accordingly (Section 3).
- The element uses a single global spacing Δx (the minimum nodal distance) to size every kernel. Clouds must therefore be quasi-uniform. Section 5.1 shows the effect of an unbalanced cloud.

2.3 Nodal integration and natural stabilization

Integrals over the mid-surface are evaluated at the nodes, $\int_{\Omega} f \, dA \approx \sum_K f(\boldsymbol{\xi}_K) A_K$, with Voronoi nodal areas A_K . Pure nodal integration is rank deficient. Following the paper (its Eq. 33 and Section 3.10), the membrane part of the internal virtual work is corrected by the first-order Taylor term of the integrand over each nodal cell,

$$\delta W_K^{\text{stab}} = \sum_{l=1}^2 \delta \boldsymbol{\varepsilon}_{,\xi^l} : \boldsymbol{\sigma}_{,\xi^l} V_K M_{Kl}, \quad V_K = A_K h, \quad M_{Kl} = \frac{A_K}{12}, \quad (3)$$

evaluated at the mid-surface. The stress gradient $\boldsymbol{\sigma}_{,\xi^l}$ is carried as a history variable and advanced with the mid-surface algorithmic tangent, $\boldsymbol{\sigma}_{,\xi^l} \leftarrow \boldsymbol{\sigma}_{,\xi^l} + \mathbb{C}^{\text{alg}} \mathbf{B}_{,\xi^l} \Delta \mathbf{d}$, then co-rotated with the material frame (the paper’s Eqs. 67–72; `stab_siggrad=1`, the default). Because the algorithmic tangent collapses along the plastic flow direction, the stabilization does not stiffen a localizing band; this is what lets the necking example (Section 5.4) soften like the paper’s stabilized result rather than locking.

Two deviations from the paper are deliberate. The cell moment uses $A_K/12$ in both tangent directions (equal to the paper’s $\Delta \xi_l^2/12$ for square cells). An optional *bending* stabilization, the curvature analogue of Eq. 33, is available as `stab_bending`; the paper implements only the membrane term. It is off by default and needed only for thin, nearly inextensional, bending-dominated problems such as the pinched hemisphere (Section 5.2).

2.4 Through-thickness plasticity, co-rotation and thickness update

Stresses are integrated at three Gauss stations through the thickness. At each station a plane-stress J_2 model with isotropic hardening

$$Y(\bar{\varepsilon}^p) = Y_0 + H \bar{\varepsilon}^p + (Y_{\text{sat}} - Y_0)(1 - e^{-\delta \bar{\varepsilon}^p}) \quad (4)$$

is integrated by a return mapping, with $\sigma_{33} = 0$ enforced by iterating on the through-thickness rate D_{33} (the paper’s Algorithm 3; `PlaneStressJ2.h`). Stress is updated in a co-rotational frame obtained from a Green–Naghdi / Flanagan–Taylor polar decomposition of the incremental deformation (Algorithm 2; `FlanaganTaylor.h`, `corotational=1`), which keeps the update objective under finite rotation. With `thickness_update=1` the thickness follows the mid-surface D_{33} through the paper’s second-order update (Eq. 79), $h_{n+1} = h_n (1 + \frac{1}{2} \Delta t D_{33}) / (1 - \frac{1}{2} \Delta t D_{33})$, which is what triggers localization in the necking example.

2.5 Dynamics

The elasto-plastic cases are integrated with Tahoe’s explicit central-difference scheme and a translational lumped mass $m_K = \rho A_K h$. The internal force pass is parallel over nodes (OpenMP) with assembly kept serial and in the original order, so results are bit-identical to the single-threaded code; on 8,480 nodes it runs about 2.5 times faster on 8 threads.

2.6 Features implemented but not yet validated

Self-contact by a pinball potential (the paper’s Section 3.13; `contact_stiffness`) and rotational-continuity penalty coupling of shell patches at a kink (Section 3.12; `penalty_coupling`) are implemented, unit-tested on their operators and demonstrated on a two-sheet contact toy problem. They are off by default and have not been validated on the paper’s multi-patch tube-crush example.

3 Essential boundary conditions on a non-interpolatory field

Because $u^h(\mathbf{x}_A) = \sum_J \Psi_J(\mathbf{x}_A) d_J \neq d_A$, an ordinary Tahoe kinematic boundary condition, which prescribes the coefficient d_A , does not prescribe the physical displacement. On a coarse pinched cylinder a coefficient displacement of 150 mm delivered only about 52 mm of physical displacement. Two treatments are available.

3.1 Direct nodal collocation

`kinematic_BC...collocation="1"` (`CollocationKBCT`, issue #70) solves, every step, for the constrained coefficients that make the physical value exact,

$$\Phi_c \mathbf{d}_c = \bar{\mathbf{u}} - \Phi_f \mathbf{d}_f, \quad (5)$$

and imposes them as ordinary prescribed values. It matches the paper’s statement that essential conditions are collocated in dynamics. It is not a variational enforcement: the free coefficients in the support of a constrained node receive no constraint force, and the reaction reported on the constrained coefficient is a generalized force, not the physical load.

3.2 Variational penalty enforcement: `penalty_displacement_meshfree`

The force controller `MFPenaltyDisplacementT` enforces the physical constraint

$$g_A = \sum_J \Psi_J(\mathbf{x}_A) d_J - \bar{u}_A(t) = 0 \quad (6)$$

for one displacement component at every node A of a node set, with the Galerkin force acting on every coefficient in the support,

$$f_J = \Psi_J(\mathbf{x}_A) \lambda_A, \quad \lambda_A = -k g_A - c_A \dot{g}_A, \quad c_A = \zeta 2\sqrt{k m_A^{\text{eff}}}, \quad \frac{1}{m_A^{\text{eff}}} = \sum_J \frac{\Psi_J(\mathbf{x}_A)^2}{m_J}. \quad (7)$$

Here k is the penalty stiffness, ζ the damping ratio (a critically damped dashpot, $\zeta = 1$, suppresses the ringing of the penalty spring in explicit runs) and m_J the lumped masses that the element reports through the `MeshFreeCollocationSupportT::NodalMass` hook. λ_A is the physical point reaction, work-conjugate to $u^h(\mathbf{x}_A)$. The controller also assembles the tangent $k \Psi \Psi^T$ and registers its equation coupling, so it works in implicit analyses: the static linear pinched cylinder converges in one Newton iteration with $\lambda = k \times \text{slack}$. A deck entry reads

```
<penalty_displacement_meshfree dof="1" schedule="1" value="-300.0"
    penalty="1.0e5" damping_ratio="1.0">
  <node_ID_list><String value="3"/></node_ID_list>
</penalty_displacement_meshfree>
```

and each output step writes one line per node with $u^h(\mathbf{x}_A)$, λ_A and their means since the previous output.

On the pinched cylinder the collocation reaction and the penalty multiplier agree within about 10% over the whole 0–300 mm crush, so the choice of treatment was not the source of the historical Fig. 18 discrepancy (Section 5.5); the penalty controller is nevertheless the correct measurement.

4 Implementation in Tahoe

Table 1: Source files.

File	Content
development/src/elements/meshfree_kl_shell/RKShellT.{h,cpp}	element: setup, charts, shape functions, internal force, stabilization, output, contact, coupling
.../KLShellKernels.h	KL auxiliary tensors, strain-displacement and curvature operators
.../PlaneStressJ2.h	plane-stress J_2 return, algorithmic tangent, Padé thickness stretch
.../FlanaganTaylor.h	co-rotational rotation and stretch update
tahoe/src/nodes/KBC_controllers/CollocationKBCT, CollocationSolverT	direct nodal collocation
tahoe/src/nodes/FBC_controllers/MFPenaltyDisplacementT	penalty displacement controller
tahoe/src/nodes/KBC_controllers/MeshFreeCollocationSupportT.h	element interface: collocation rows $\Psi_J(\mathbf{x}_A)$ and nodal masses

The element derives from `ElementBaseT`; it is not built on Tahoe’s `MeshFreeSupportT` or the EFG/RKPM solid elements, because the three-dimensional solid meshfree path rejects a surface point cloud. Only the shape-function core is shared. Element parameters are listed in `docs/meshfree_kl_shell/README.md` and in the Tahoe User Guide. The defaults reproduce the paper’s formulation (`kernel=1`, `stab_natural=1`, `stab_siggrad=1`, `corotational=1`) except `completeness`, which defaults to cubic; the paper’s quadratic basis needs `completeness="2"`, as in all decks of Section 5.

5 Verification and validation

All results in this section were produced with the current `develop` code (September 2026). Deflections are physical, $u^h(\mathbf{x}_A)$, not coefficients.

Table 2: Summary against the reference paper.

Case (paper section)	Reference	Result
Scordelis–Lo roof (4.2.1)	0.3006; paper Fig. 7	0.259 / 0.290 / 0.298 / 0.300 vs paper 0.261 / 0.290 / 0.29
Pinched hemisphere (4.2.2)	0.0924	0.0207 / 0.0787 / 0.0909 vs paper 0.0105 / 0.065 / 0.090
Linear pinched cylinder (4.2.3)	1.8248e-5	0.593 / 1.290 / 1.622 / 1.745 vs paper 1.055 / 1.627 / 1.78
Necking cylinder (4.3)	paper Fig. 15	peak 614 vs 609 MPa; within 1–4% to $U_{\text{norm}} = 10.7$ mm
Elasto-plastic pinched cylinder (4.4)	paper Fig. 18	inside the published band with the $F/4$ convention

5.1 Scordelis–Lo roof

Cylindrical roof, $R = 25$, $L = 50$, $h = 0.25$, 80° span, $E = 4.32 \times 10^8$, $\nu = 0$, gravity load 90 per unit area, rigid diaphragms, normalized support 3.0. Vertical deflection at the midpoint of the free edge, on the paper’s four quasi-uniform clouds:

The reference value is 0.3006. The regression deck previously used a 225-node cloud whose spacing was 2.5 across the arc but 3.6 along the axis. With one global spacing sizing the kernel, the support barely reached the axial neighbours and the deflection sat at 0.194. On the paper’s balanced clouds the result reproduces the paper’s own convergence to three digits.

Nodes	Cloud	Tahoe	Paper Fig. 7(b)
315	15×21	0.2591	0.261
1,189	29×41	0.2897	0.290
4,617	57×81	0.2978	0.298
18,193	113×161	0.2999	0.300

5.2 Pinched hemisphere

Hemisphere of radius 10 and thickness 0.04 ($R/h = 250$; the paper prints 0.4, which is inconsistent with the reference deflection), $E = 6.825 \times 10^7$, $\nu = 0.3$, free equator, two pairs of opposite radial point loads, normalized support 3.0. Radial displacement at a load point:

Nodes per side	Nodes (full model)	Tahoe	Paper Fig. 9(b)
11	277	0.0207	0.0105
21	1,061	0.0787	0.065
41	4,157	0.0909	0.090

The reference is 0.0924; at 41 nodes per side the result is within 1.6% of it and of the paper. Two choices differ from the paper. The whole hemisphere is modelled, because a quarter model needs rotation conditions on the symmetry planes that a rotation-free shell cannot impose directly. And the optional bending stabilization is switched on: with the membrane stabilization of Eq. 33 alone the stiffness is singular on this thin, bending-dominated shell (a result that depends on the linear solver). The two coarser clouds give larger deflections than the paper's; the cause has not been investigated. The 81-node-per-side cloud (16,469 nodes) was not rerun: the element's setup did not finish within 30 minutes at that size.

5.3 Linear pinched cylinder

Cylinder $R = 300$, $L = 600$, $h = 3$, $E = 3 \times 10^6$, $\nu = 0.3$, rigid diaphragms, two opposite unit point loads at mid-length, normalized support 2.5, the paper's full-cylinder clouds. Radial displacement under the load, in units of 10^{-5} :

Nodes	Cloud	Tahoe	Paper Fig. 11(b)	Tahoe / reference
330	30×11	0.593	1.055	0.32
1,260	60×21	1.290	1.627	0.71
4,920	120×41	1.622	1.783	0.89
19,440	240×81	1.745	1.822	0.96

The reference is 1.8248×10^{-5} (the paper's text states 1.8624×10^{-5} , but its Fig. 11(b) plots 1.825). The element converges to the reference but more slowly than the paper's RKPM result, which is already within 2% at 4,920 nodes; the finest cloud is 4% stiff. This is the membrane-locking behaviour listed in Section 7: on this inextensional bending problem the coarse clouds carry spurious membrane strain.

5.4 Necking of a cylindrical shell

Cylinder $R = 10$ mm, $L = 50$ mm, $h = 1$ mm; $E = 189$ GPa, $\nu = 0.29$, $Y = 343 + 300 \bar{\varepsilon}^p + 337(1 - e^{-16.93 \bar{\varepsilon}^p})$ MPa; one end fixed, the other pulled axially with a velocity ramped to 30 m/s over 0.5 ms; $\rho = 7800$ kg/m³, $\Delta t = 4 \times 10^{-8}$ s; normalized support 2.4; thickness update on. The cloud has 8,262 nodes (the paper's has 8,284). The effective stress is the axial reaction over the undeformed section, $|R_z|/(2\pi Rh)$, plotted against the nodal displacement norm $U_{\text{norm}} = (\sum_I \|\mathbf{u}_I\|^2/N)^{1/2}$ as in the paper.

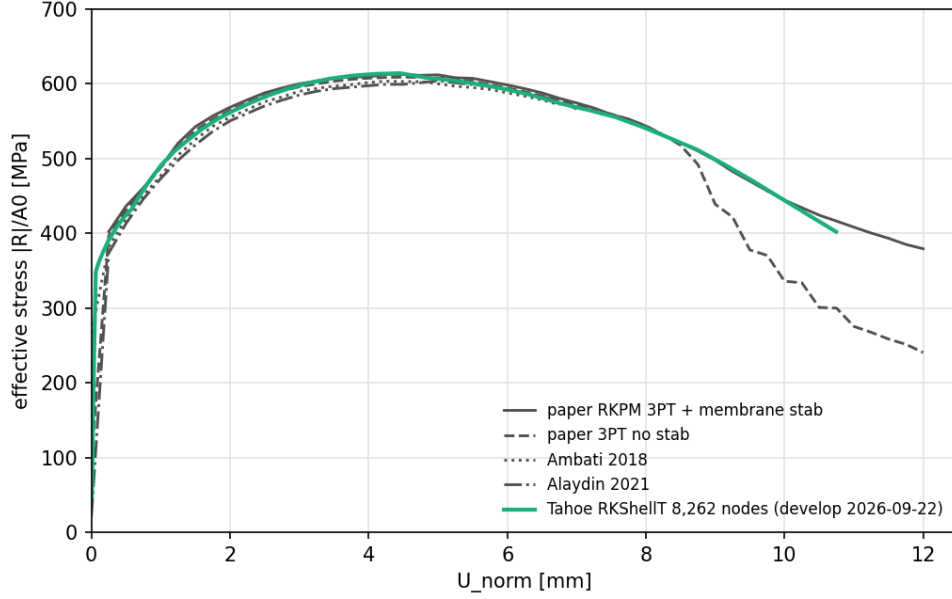


Figure 1: Necking: Tahoe against the digitized paper Fig. 15.

U_{norm} [mm]	Paper, 3 points + stabilization	Ambati et al. [2]	Tahoe
1	489	477	492
2	569	555	564
4	609	602	613
6	598	588	593
8	543	—	542
10	444	—	447

The run follows the paper’s stabilized curve from the elastic rise through the peak (614 against 609 MPa) and down the softening branch to $U_{\text{norm}} = 10.7$ mm, clearly separated from the paper’s unstabilized branch, which is near 300 MPa there. Beyond $U_{\text{norm}} \approx 10.9$ mm the explicit run becomes unstable in the neck (Section 7).

5.5 Elasto-plastic pinched cylinder

Cylinder $R = 300$ mm, $L = 600$ mm, $h = 3$ mm, rigid diaphragms; $E = 3000$ MPa, $\nu = 0.3$, $Y = 24.3 + 300 \varepsilon^p$ MPa; the two diametrically opposite mid-length points are driven inward to 300 mm by the penalty controller, and the reported load is its multiplier λ . Mass-scaled explicit dynamics, 50,000 steps, quasi-static past about 60 mm (kinetic energy below 5% of the external work).

The load convention. Every curve in the paper’s Fig. 18 comes from, or matches, a symmetry model: Ambati et al. [2] model one quarter of the shell and Alaydin et al. [3] one octant. In such a model the load point lies on two symmetry planes, so the applied nodal force, which is what is plotted as the reaction, is one quarter of the physical pinch load. For a long time this implementation appeared 4–9 times too stiff because its physical load was compared with those curves directly.

On the paper’s resolution (160 by 53 nodes) the shell lies inside the band of the published results over the whole crush and reproduces the load dip near 160 mm. An independent reference built from Tahoe’s finite-strain solid elements (one-eighth model, 20-node hexahedra, Simo J_2) agrees with the shell to 100 mm and reproduces the linear obstacle-course deflection to 0.5%. On

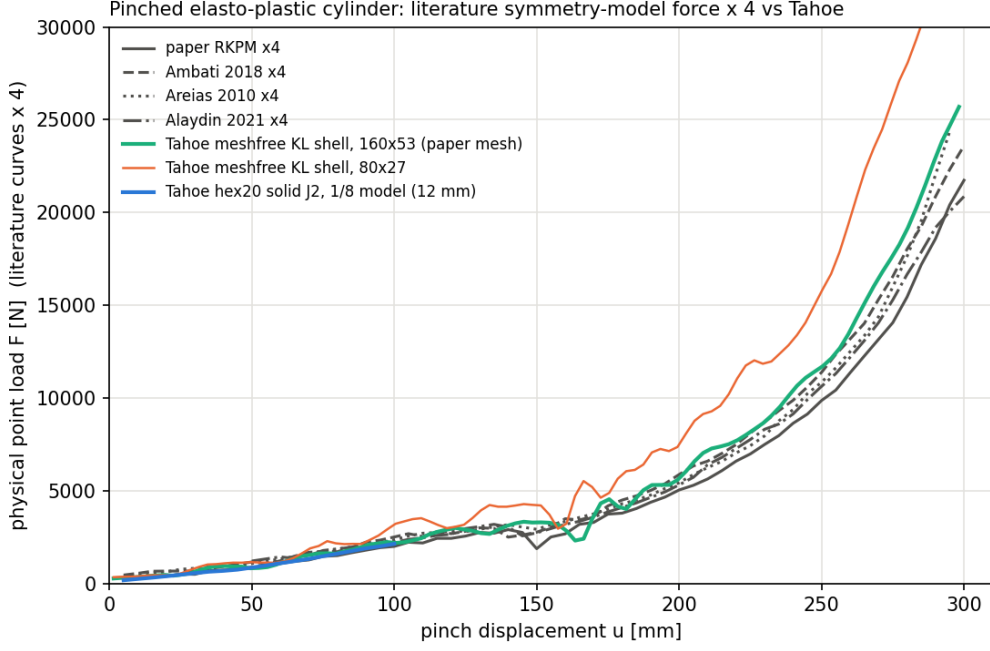


Figure 2: Elasto-plastic pinched cylinder: the published curves multiplied by 4 against the physical point load from Tahoe’s meshfree shell (two clouds) and an independent Tahoe hex20 solid model.

u [mm]	Paper $\times 4$	Ambati $\times 4$	Areias $\times 4$	Shell, 8,480 nodes	Solid, hex20
50	874	1156	1069	814	852
100	1995	2502	2356	2182	2131
200	5036	5870	5425	5580	—
300	21733	23595	—	25692	—

the way to this result the following were tested at the paper’s resolution and excluded as causes of the apparent mismatch: the boundary-condition treatment and the reported force, mesh resolution (a 33,600-node cloud agrees with 8,480 nodes within 5%), the Taylor stabilization, and the end conditions. Coarser clouds remain visibly too stiff (Fig. 2, 80 by 27 nodes).

6 Tests, benchmarks and continuous integration

Unit tests (`tests/meshfree/`, Google Test, built with `TAHOE_DEV=ON`): shape functions on the chart (partition of unity, reproduction, derivative consistency), the kinematic chain against finite differences, assembly (symmetry, rigid null space, patch tests), stress update, plane-stress J_2 return, Scordelis–Lo through the in-memory element and through the `tahoe` binary, the thickness update and co-rotation, and the collocation transform.

Regression benchmarks (`benchmark_XML`, compared with the `compare` tool):

- `level.0/meshfree_kl_shell`: Scordelis–Lo on the paper’s 315-node cloud, the hemisphere and linear pinched cylinder, the collocation boundary condition, and the penalty displacement controller on the explicit and implicit paths.
- `level.2/meshfree_kl_shell`: shortened, mass-scaled versions of the necking cylinder and the elasto-plastic pinched cylinder (about 45 s each on one thread).

The validation decks and scripts behind Section 5 are in `docs/meshfree_kl_shell/fig18_validation` and `necking_validation`.

Continuous integration and solver verification. Three GitHub Actions workflows build and test every push and pull request to `develop` and `master`: a serial build (SPOOLES and SuperLU), a MUMPS build, and an MPI build. Each runs the Google Test suite; the serial and MUMPS workflows then run benchmark levels 0–2 and the MPI workflow level 3 on four ranks. A benchmark step fails only on a crash or timeout; comparison failures are reported but tolerated, because level 0 carries a known set of legacy failures (issue #37).

From 21 to 25 September 2026 all three workflows failed on one unit test: the pipeline test ran a Scordelis–Lo deck from the untracked personal `applications/` directory, which does not exist on a CI runner. The test now runs the tracked level 0 deck. Before merging, the four configurations below were built from scratch and tested locally; the first three reproduce the CI workflows.

Table 3: Pre-merge verification, 25 September 2026.

Configuration	Unit tests	Level 0	Levels 1 / 2	I
Serial (SPOOLES, SuperLU)	95 / 95	173 pass, 0 crash	103 / 103, 49 / 49	–
MUMPS	95 / 95	173 pass, 0 crash	103 / 103, 49 / 49	–
MPI	95 / 95	–	–	2
All solvers (MPI, MUMPS, METIS, SPOOLES-MT, SuperLU)	95 / 95	–	–	2

The benchmark references cover the default solvers only, so the other solvers were cross-validated: the same problem was solved with every solver and compared with SPOOLES. The problems were an 18,000-degree-of-freedom nonlinear electro-mechanical liquid inclusion (27 Newton iterations), a large-strain J_2 patch (6 iterations, 224 nodal values) and a linear diffusion bar. SuperLU, serial MUMPS, SPOOLES-MT with 4 threads, and SPOOLES-MPI and MUMPS-MPI on 2 and 4 ranks, each with METIS and with the built-in bisection partitioner, all reproduce the SPOOLES Newton history (identical iteration counts, residuals within 2×10^{-6} relative) and its nodal output to machine precision ($< 10^{-15}$ relative). Two exceptions were found. The profile (skyline) solver is correct but too slow for the 18,000-degree-of-freedom problem. SPOOLES-MPI fails when the 10-node diffusion bar is split over four ranks, one of which owns a single equation; MUMPS-MPI solves the same partition correctly.

One regression deck, `level.1/.../material.120/wlc_superlu.xml`, aborts at step 11 of 290 with every solver, identically before and after these merges, yet is counted as passing: it has no stored reference and `tahoe` returns exit code 0 after a solver exception. Checking the run log for exceptions in `run_benchmarks.sh` would expose such cases. (Follow-up, issues #74 and #82: the runner now reports aborted runs, and the abort itself was a boundary-condition transcription error that left the x translation free; with it corrected the deck runs all 290 steps.)

7 Limitations and outlook

- **Deep necking.** The explicit necking run becomes unstable at $U_{\text{norm}} \approx 10.9$ mm, with a plastic strain of 1.28 and the thickness at 45% of its initial value in a 3.75 mm band; the paper continues to 12.2 mm and strains near 2. Candidates are the stable time step at reduced thickness, the Padé thickness update near its pole, and the scaling of the stabilization with the current thickness.
- **Membrane locking** (issue #68). Coarse clouds are too stiff on inextensional bending (the 80 by 27 elasto-plastic pinched cylinder; the linear pinched cylinder at 0.71 of the reference on 1,260 nodes where the paper reaches 0.89). The paper’s Section 5 sketches remedies; a

patch-test-preserving stress projection in the spirit of [4] is the classical alternative. Two naive lower-order membrane projections were tried and did not soften the response.

- **Implicit plasticity.** Static elasto-plastic runs use the elastic stiffness and slow down once yielding spreads. A consistent or numerical tangent would make quasi-static shell plasticity cheap and remove the reliance on mass-scaled explicit runs.
- **Cubic completeness** reduces locking statically but is unstable with the lumped mass in explicit dynamics.
- **Quasi-uniform clouds.** One global spacing sizes all kernels; locally refined clouds need per-node support sizes.
- **Bending stabilization** is required for the thin hemisphere and is not part of the paper’s formulation; self-contact and patch coupling are not yet validated on a full problem.

References

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